

Activity: Build Oracle JET Application using Jenkins Pipeline

Objective

In this activity, you will configure a Jenkins Pipeline to automatically download an Oracle JET application from GitHub, install the required dependencies, and build the project.

Prerequisites

- Jenkins installed and running
- Git installed
- Node.js and npm installed
- Oracle JET CLI installed
- GitHub account
- Oracle JET project (**my-i18n-app**) pushed to GitHub

Project Details

Item	Value
Project Name	my-i18n-app
Repository	<code>https://github.com/<your-username>/ojet-jenkins-demo.git</code>
Branch	main

Activity Steps

Step 1: Open Jenkins

Open a web browser and navigate to:

`http://<jenkins-server>:8080`

Login using your Jenkins credentials.

Step 2: Create a New Pipeline

1. Click **New Item**.
2. Enter the job name:

OJET-Form-Pipeline

3. Select **Pipeline**.
4. Click **OK**.

Step 3: Configure the Pipeline

Scroll to the **Pipeline** section.

Choose:

Definition:
Pipeline Script

Step 4: Add the Pipeline Script

Copy and paste the following Jenkins Pipeline:

```
pipeline {  
    agent any  
  
    stages {  
  
        stage('Checkout') {  
            steps {  
                git branch: 'main',  
                    url: 'https://github.com/arora026richa/ojet-jenkins-  
demo.git'  
            }  
        }  
  
        stage('Install Dependencies') {  
            steps {  
                sh 'npm install'  
            }  
        }  
    }  
}
```

```
    stage('Build Oracle JET') {  
  
        steps {  
  
            sh 'npx ojet build'  
  
        }  
  
    }  
  
    stage('Build Complete') {  
  
        steps {  
  
            echo 'Oracle JET project built successfully!'  
  
        }  
  
    }  
  
}
```

Replace the repository URL with your own GitHub repository.

Click **Save**.

Step 5: Run the Pipeline

Click:

Build Now

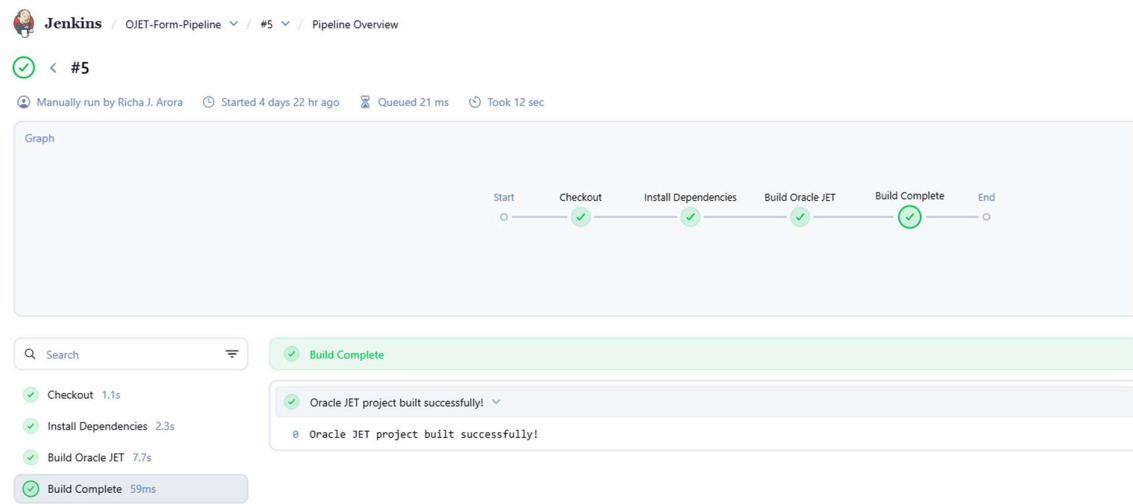
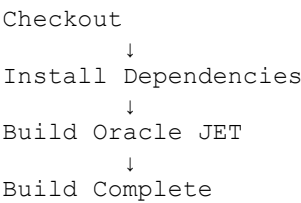
Step 6: View the Build Output

Open:

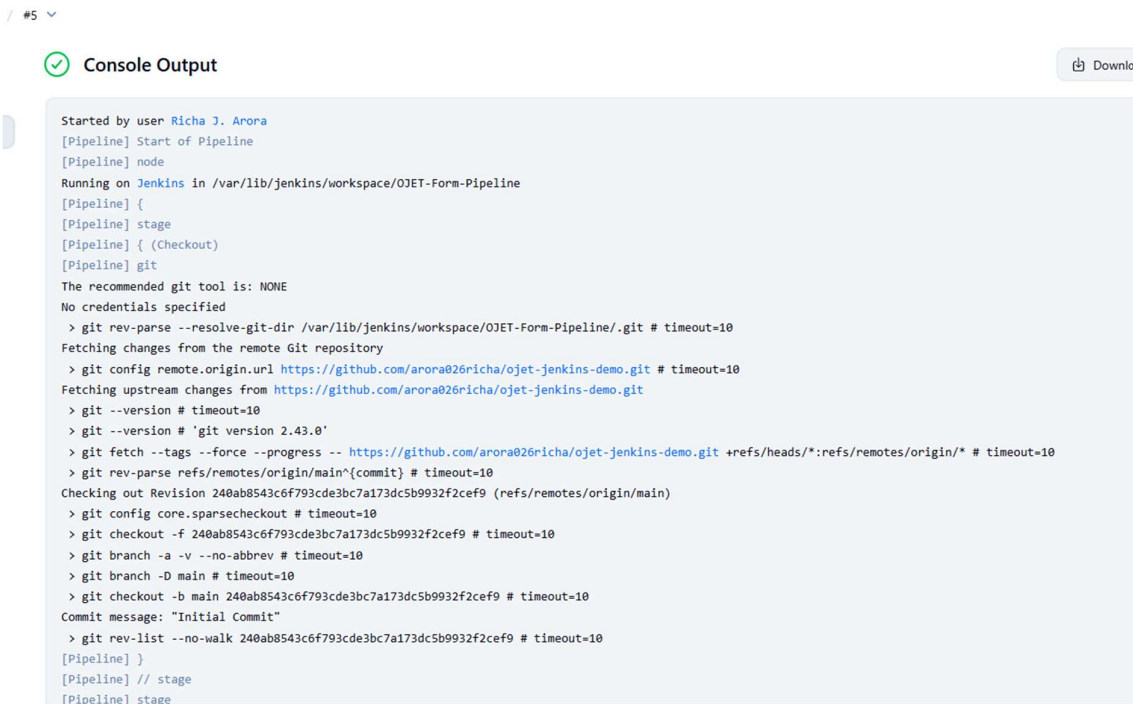
```
Build History  
  ↓  
Latest Build  
  ↓  
Console Output
```

Observe the execution of each pipeline stage.

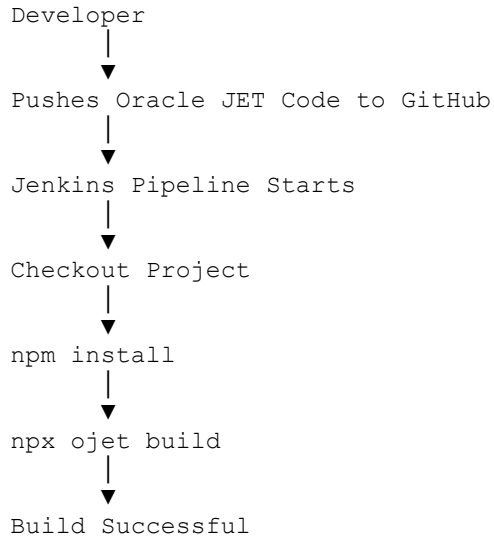
Pipeline Stages



Result



Pipeline Workflow



Understanding from this project:

Before Jenkins

Suppose a developer develops a new feature in an Oracle JET or CRUD application.

Without Jenkins, the developer has to manually:

- Download the latest code.
- Install all project dependencies.
- Build the application.
- Check whether it works.
- Deploy it to the server.

Imagine 20 developers doing this every day. It is time-consuming and error-prone.

This is where **Jenkins automates the entire process.**

What Happens After Running the Pipeline?

1. Checkout

Jenkins downloads the latest version of the project from the GitHub repository into its workspace. This ensures that the build always uses the most recent source code.

2. Install Dependencies

Jenkins executes `npm install` to download and install all the libraries required by the application, as specified in the `package.json` file.

3. Build Application

Jenkins builds the application and checks for any compilation or syntax errors. If the build is successful, the application is ready for deployment.

4. Deploy Application

The built application is automatically deployed to the target environment, making the latest version available for users.

5. Build Status

Finally, Jenkins displays the build result. A **SUCCESS** status indicates that all stages completed successfully, while a **FAILURE** status helps developers identify and fix issues before the application reaches users.